

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404014
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Allen; Jason Bradley

Hydraulic motor start assist

Abstract

A nose-wheel steering system is provided. The nose-wheel steering system includes a hydraulic motor including a drive shaft configured to rotate about a first axis; and a hydraulic motor start assist mechanism coupled to the hydraulic motor and configured to provide a temporary boost of hydraulic fluid pressure supplied to the hydraulic motor.

Inventors:	Allen; Jason Bradley (Waco, TX)
Applicant:	GOODRICH CORPORATION (Charlotte, NC)
Family ID:	90368025
Assignee:	GOODRICH CORPORATION (Charlotte, NC)
Appl. No.:	18/187416
Filed:	March 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240317390 A1	Sep. 26, 2024

Publication Classification

Int. Cl.: **B64C25/50** (20060101); **B64C25/22** (20060101)

U.S. Cl.:

CPC **B64C25/50** (20130101); **B64C25/22** (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2001620	12/1934	Lucien	244/102R	B64C 25/22
2267284	12/1940	Livers	60/904	F15B 13/021
2363235	12/1943	Ellinwood	137/637.1	B64C 25/22
2484603	12/1948	Audemar	416/157R	B64C 25/22
4190130	12/1979	Beck	180/406	B62D 5/32
4422290	12/1982	Huffman	60/404	F15B 11/17
4574904	12/1985	Goode	60/404	B62D 5/32
4694649	12/1986	Howeth	N/A	N/A
6345501	12/2001	McKay	60/494	F15B 21/14
6792844	12/2003	Gedge	251/63.4	F15B 13/021
8393565	12/2012	Frank	244/50	B64C 25/22
9422052	12/2015	Kondo	N/A	B64C 13/504
10196131	12/2018	McCormick et al.	N/A	N/A
10464663	12/2018	Luce	N/A	B64C 25/50
10794370	12/2019	Abaitancei	N/A	B60K 6/12
10851815	12/2019	Frank	N/A	B64C 25/30
10919620	12/2020	Didey	N/A	B64C 25/34
11180244	12/2020	Evans	N/A	F15B 15/06
11260961	12/2021	Howell	N/A	B64C 25/42
11820495	12/2022	Kerr	N/A	B64C 25/30
2003/0188529	12/2002	Collet et al.	N/A	N/A
2005/0082427	12/2004	Seung	244/102R	B64C 25/50
2006/0144993	12/2005	Hsu et al.	N/A	N/A
2010/0301170	12/2009	Boseroy	244/194	G05B 11/42
2019/0120258	12/2018	Zavadinka	N/A	N/A
2022/0281591	12/2021	Allen	N/A	N/A
2024/0317390	12/2023	Allen	N/A	B64C 25/50

OTHER PUBLICATIONS

European Patent Office, European Search Report dated Aug. 21, 2024 in Application No. 24164725.4. cited by applicant

Background/Summary

FIELD

(1) The present disclosure relates generally to aircraft steering systems and, more particularly, to a hydraulic motor start assist for aircraft nose-wheel steering systems.

BACKGROUND

(2) Aircraft typically employ nose-wheel steering systems to steer the aircraft while taxiing on the ground. A typical nose-wheel steering system includes a collar gear provided upon a strut associated with the nose-wheel. Various actuators and gear trains may be associated with rotating the collar gear, and hence the strut, thereby adjusting the orientation of the nose-wheel to affect steering (i.e., the direction of the taxiing aircraft). Hydraulic motor sizing plays a large role in the weight of a rotary actuator. While nominal running torque efficiencies of 90-95% are typical of a hydraulic motor, the minimum starting torque efficiency may be as low as roughly 65-70%. Therefore, since actuator motor torque load may be at maximum at start, the motor is typically oversized to compensate for this lower starting torque efficiency, leading to more weight and package volume.

SUMMARY

(3) Disclosed herein is a nose-wheel steering system. The nose-wheel steering system includes a hydraulic motor including a drive shaft configured to rotate about a first axis; and a hydraulic motor start assist mechanism coupled to the hydraulic motor and configured to provide a temporary boost of hydraulic fluid pressure supplied to the hydraulic motor.

(4) In various embodiments, the hydraulic motor start assist mechanism includes a servo valve coupled to the hydraulic motor and an intensifier valve, the servo valve configured to provide the hydraulic fluid pressure to the hydraulic motor and the intensifier valve; the intensifier valve coupled to a displacement intensifier and configured to provide the hydraulic fluid pressure to the servo valve in a first state and provide the hydraulic fluid pressure to the displacement intensifier in a second state; and the displacement intensifier configured to translate in a first direction in the first state and translate in a second direction in the second state.

(5) In various embodiments, responsive to receiving a first command from a pilot steering input, the servo valve switches from a flow-blocking position to a cross-flow position. In various embodiments, in the cross-flow position, the servo valve provides pressurized hydraulic fluid to a first port on the hydraulic motor. In various embodiments, responsive to the first command being discontinued, the servo valve switches from the cross-flow position to the flow-blocking position. In various embodiments, responsive to the hydraulic fluid pressure at the first port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, a rise in hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the first port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor. In various embodiments, an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in the hydraulic fluid pressure at the first port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement intensifier. In various embodiments, the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by the area at the head end the displacement intensifier being greater than an area at a tail end of the displacement intensifier thereby allowing a piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at a

back end of the displacement intensifier to increase in pressure.

(6) In various embodiments, responsive to receiving a second command from a pilot steering input, the servo valve switches from a flow-blocking position to a straight-flow position. In various embodiments, in the straight-flow position, the servo valve provides pressurized hydraulic fluid to a second port on the hydraulic motor. In various embodiments, responsive to the second command being discontinued, the servo valve switches from the straight-flow position to the flow-blocking position. In various embodiments, responsive to the hydraulic fluid pressure at the second port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, a rise in the hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the second port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor. In various embodiments, an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in the hydraulic fluid pressure at the first port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement intensifier. In various embodiments, the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by at the area at the head end of the displacement intensifier being greater than an area at a tail end of the displacement intensifier thereby allowing a piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at a back end of the displacement intensifier to increase in pressure.

(7) Also disclosed herein is a nose landing gear assembly. The nose landing gear assembly includes a shock strut assembly including a strut cylinder and a strut piston configured to telescope relative to the strut cylinder; and a nose-wheel steering system coupled to the shock strut assembly and configured to rotate the strut piston about a piston axis of rotation. The nose-wheel steering system including a hydraulic motor including a drive shaft configured to rotate about a second axis, the second axis being parallel to the piston axis of rotation; and a hydraulic motor start assist mechanism coupled to the hydraulic motor and configured to provide a temporary boost of hydraulic fluid pressure supplied to the hydraulic motor.

(8) In various embodiments, the hydraulic motor start assist mechanism includes a servo valve coupled to the hydraulic motor and an intensifier valve, the servo valve configured to provide the hydraulic fluid pressure to the hydraulic motor and the intensifier valve; the intensifier valve coupled to a displacement intensifier and configured to provide the hydraulic fluid pressure to the servo valve in a first state and provide the hydraulic fluid pressure to the displacement intensifier in a second state; and the displacement intensifier configured to translate in a first direction in the first state and translate in a second direction in the second state.

(9) In various embodiments, responsive to receiving a first command from a pilot steering input, the servo valve switches from a flow-blocking position to a cross-flow position. In various embodiments, in the cross-flow position, the servo valve provides pressurized hydraulic fluid to a first port on the hydraulic motor. In various embodiments, responsive to the first command being discontinued, the servo valve switches from the cross-flow position to the flow-blocking position. In various embodiments, responsive to the hydraulic fluid pressure at the first port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, the hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the first port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor. In various embodiments, an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in the hydraulic fluid pressure at the first port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement

intensifier. In various embodiments, the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by the area at the head end of the displacement intensifier being greater than an area at a tail end of the displacement intensifier thereby allowing a piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at the back end of the displacement intensifier to increase in pressure.

(10) In various embodiments, responsive to receiving a second command from a pilot steering input, the servo valve switches from a flow-blocking position to a straight-flow position. In various embodiments, in the straight-flow position, the servo valve provides pressurized hydraulic fluid to a second port on the hydraulic motor. In various embodiments, responsive to the second command being discontinued, the servo valve switches from the straight-flow position to the flow-blocking position. In various embodiments, responsive to the hydraulic fluid pressure at the second port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, the hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the second port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor. In various embodiments, an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in the hydraulic fluid pressure at the second port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement intensifier. In various embodiments, the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by the area at the head end of the displacement intensifier being greater than an area at a tail end of the displacement intensifier thereby allowing a piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at the back end of the displacement intensifier to increase in pressure.

(11) Also disclosed herein is a shock strut assembly for an aircraft landing gear assembly. The aircraft landing gear assembly includes a strut cylinder; a strut piston configured to telescope relative to the strut cylinder; and a steering system coupled to the strut piston and configured to rotate the strut piston about a piston axis of rotation. The steering system includes a hydraulic motor including a drive shaft configured to rotate about a second axis, the second axis being parallel to the piston axis of rotation; and a hydraulic motor start assist mechanism coupled to the hydraulic motor and configured to provide a temporary boost of hydraulic fluid pressure supplied to the hydraulic motor.

(12) In various embodiments, the hydraulic motor start assist mechanism includes a servo valve coupled to the hydraulic motor and an intensifier valve, the servo valve configured to provide the hydraulic fluid pressure to the hydraulic motor and the intensifier valve; the intensifier valve coupled to a displacement intensifier and configured to provide the hydraulic fluid pressure to the servo valve in a first state and provide the hydraulic fluid pressure to the displacement intensifier in a second state; and the displacement intensifier configured to translate in a first direction in the first state and translate in a second direction in the second state.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the detailed description and claims when considered in connection with the drawing figures, wherein like numerals denote like elements.

(2) FIG. 1 illustrates an aircraft having left, right and nose landing gear assemblies and wheels mounted thereon, in accordance with various embodiments.

- (3) FIG. 2 illustrates a nose landing gear assembly, in accordance with various embodiments.
- (4) FIGS. 3A, 3B, 3C, 3D, and 3E illustrate a hydraulic motor start assist mechanism for use with a hydraulic motor for a nose-wheel steering system, in accordance with various embodiments.

DETAILED DESCRIPTION

(5) The detailed description of embodiments herein makes reference to the accompanying drawings, which show embodiments by way of illustration. While these embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that logical, chemical, and mechanical changes may be made without departing from the spirit and scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not for limitation. For example, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option. Further, any steps in a method discussed herein may be performed in any suitable order or combination.

(6) Disclosed herein is a hydraulic motor start assist mechanism. In various embodiments, the hydraulic motor start assist mechanism utilizes an integral intensifier and intensifier valve to temporarily boost hydraulic fluid pressure supplied to a bi-directional hydraulic motor, therefore increasing motor torque output by roughly 30% to compensate for the lower starting torque efficiency. In various embodiments, the bi-directional hydraulic motor may be sized approximately 20-30% smaller compared to what the bi-directional hydraulic motor would normally be sized at without the hydraulic motor start assist mechanism. In various embodiments, the bi-directional hydraulic motor weight savings of approximately 25% is reduced to approximately 15-20% by the added weight of the integral intensifier and intensifier valve of the hydraulic motor start assist mechanism.

(7) With reference to FIG. 1, an aircraft **100** is illustrated. In accordance with various embodiments, aircraft **100** may include one or more landing gear assemblies, such as, for example, a left landing gear assembly **102** (or port-side landing gear assembly), a right landing gear assembly **104** (or starboard-side landing gear assembly) and a nose landing gear assembly **106**. Each of left landing gear assembly **102**, right landing gear assembly **104**, and nose landing gear assembly **106** may support the aircraft **100** when not flying, allowing aircraft **100** to taxi, takeoff, and land safely and without damage to aircraft **100**. In various embodiments, left landing gear assembly **102** may include a left shock strut assembly **108** and a left wheel assembly **110**, right landing gear assembly **104** may include a right shock strut assembly **112** and a right wheel assembly **114**, and nose landing gear assembly **106** may include a nose shock strut assembly **116** and a nose wheel assembly **118**. One or more pilot steering input(s) **120** (e.g., steering wheels, pedals, knobs, or the like) may be located in a cockpit of aircraft **100**.

(8) Referring now to FIG. 2, nose landing gear assembly **106** is illustrated. In accordance with various embodiments, the nose shock strut assembly **116** of nose landing gear assembly **106** includes a strut cylinder **202** and a strut piston **204**. Strut piston **204** may be operatively coupled to strut cylinder **202**. Strut cylinder **202** may be configured to receive strut piston **204** in a manner that allows the two components to telescope with respect to one another. Strut piston **204** may translate into and out the strut cylinder **202**, thereby absorbing and damping loads imposed on nose landing gear assembly **106**. An axle **206** of nose wheel assembly **118** may be coupled to an end of strut piston **204** that is opposite strut cylinder **202**. The nose wheels have been removed from nose wheel assembly **118** in FIG. 2 to more clearly illustrate the features of the nose shock strut assembly **116**.

(9) In various embodiments, nose landing gear assembly **106** may include a torque link **208** coupled to the nose shock strut assembly **116** and/or to axle **206**. Torque link **208** includes a first (or upper) arm **210** and a second (or lower) arm **212**. First arm **210** is pivotably coupled to second arm **212**. Strut cylinder **202** is coupled to an attachment linkage **214** configured to secure the nose

shock strut assembly **116** to the aircraft **100** and to translate nose landing gear assembly **106** between the landing gear up and landing gear down positions. Nose landing gear assembly **106** may include one or more drag brace(s) such as drag brace **216**. In various embodiments, drag brace **216** may be located proximate an aft side of the nose shock strut assembly **116**. Nose landing gear assembly **106** may include one or more hydraulic fluid lines (i.e. conduits).

(10) In accordance with various embodiments, nose landing gear assembly **106** includes a hydraulic motor **250**. Hydraulic motor **250** is operably coupled to strut cylinder **202** via the nose shock strut assembly **116**. In this regard, and as described in further detail below, hydraulic motor **250** is configured to rotate strut piston **204** about a piston axis of rotation A (also reference to as “axis A”), thereby adjusting the orientation of the nose wheel assembly **118** and the taxiing direction of the aircraft **100**. Axis of rotation A may be parallel to the direction of translation of strut piston **204** relative to strut cylinder **202**.

(11) Referring now to FIG. 3A, in various embodiments, a control valve assembly **328** coupled to the hydraulic motor **250** may include a solenoid-operated directional control (or servo) valve **302**, an intensifier valve **304**, and a displacement intensifier **306**, among other valves or combination of valves suitable for controlling the flow volume and direction of flow to and from port **311** and port **313**. Control valve assembly **328** is operably coupled to a steering controller **330**. Actuation of control valve assembly **328** may be controlled via steering controller **330**. Stated differently, steering controller **330** is configured to control the opening and closing (i.e., actuation) of control valve assembly **328**, thereby controlling the flow of hydraulic fluid to and from each of port **311** and port **313**. Steering controller **330** is operably coupled to pilot steering input **120**. Steering controller **330** may send actuation commands to control valve assembly **328** based on signals received from pilot steering input **120**.

(12) In various embodiments, in operation, and with additional reference to FIG. 3B, when a command is received from pilot steering input **120** via steering controller **330** to steer the nose landing gear assembly **106** to, for example, the right 20 degrees, 20 degrees per second, the command causes the solenoid-operated directional control (or servo) valve **302** to shift from a center flow-blocking position to a cross-flow position so that hydraulic fluid, flowing from the hydraulic fluid flow source **332** via third conduit **334**, flows through intensifier valve **304** (in a center straight-flow position) to the solenoid-operated directional control (or servo) valve **302** via conduit **308**. In that regard, the intensifier valve **304** is fluidly coupled to the hydraulic fluid flow source **332** via third conduit **334** and the intensifier valve **304** is fluidly coupled to the solenoid-operated directional control (or servo) valve **302** via conduit **308**. It is noted that the hydraulic fluid also flows from intensifier valve **304** to a back-end side **320** of the displacement intensifier **306** via conduit **309** causing the piston **318** to translate in a first direction toward a head end **314** of the displacement intensifier **306** if not already bottomed on the head-end **314** of the displacement intensifier's **306** cylinder. In that regard, the intensifier valve **304** is fluidly coupled to the back-end side **320** of the displacement intensifier **306** via conduit **309**.

(13) Accordingly, in various embodiments, the hydraulic fluid then flows through solenoid-operated directional control (or servo) valve **302**, which is the cross-flow position, to port **311** of the hydraulic motor **250** via conduit **310**, the conduit **310** fluidly coupling the solenoid-operated directional control (or servo) valve **302** to port **311** of the hydraulic motor **250**. In various embodiments, hydraulic fluid returning from port **313** of the hydraulic motor **250** flows to solenoid-operated directional control (or servo) valve **302** via conduit **326**, which fluidly couples the port **313** of the hydraulic motor **250** to the solenoid-operated directional control (or servo) valve **302**. The solenoid-operated directional control (or servo) valve **302**, which is still in the cross-flow position, allows the hydraulic fluid to flow to the hydraulic fluid flow source **332** via fourth conduit **336**, which fluidly couples the solenoid-operated directional control (or servo) valve **302** to the hydraulic fluid flow source **332**.

(14) However, in various embodiments, the pressure of the hydraulic fluid, for example 2500 psi

(172.3689 bar), delivered to the hydraulic motor **250** may not be enough to overcome a starting torque needed for the hydraulic motor **250** to rotate, for example 3500 psi (241.3165 bar). In various embodiments, in order to provide the necessary hydraulic fluid pressure to the hydraulic motor **250**, displacement intensifier **306** is engaged. In various embodiments, the hydraulic fluid provided to the hydraulic motor **250** is also provided to the intensifier valve **304** via conduit **312**, which is fluidly coupled to conduit **310**. In that regard, the solenoid-operated directional control (or servo) valve **302** is fluidly coupled to the control of the intensifier valve **304** via conduit **312**. In various embodiments, with reference to FIG. 3C, while the hydraulic fluid pressure is not enough to overcome the starting torque needed for the hydraulic motor **250** to rotate, the same hydraulic fluid pressure is enough to cause intensifier valve **304** to shift to the right, i.e. from a straight-flow position, which provides for hydraulic fluid to flow from the hydraulic fluid flow source **332** via third conduit **334** through intensifier valve **304** (in an displacement-flow position) to a head end side **314** of the displacement intensifier **306** via conduit **316**. In that regard, the intensifier valve **304** (in a displacement-flow position) is fluidly coupled to the head-end side **314** of the displacement intensifier **306** via conduit **316**.

(15) In various embodiments, a force is generated to the head of piston **318** that is proportional to the area of the head of the piston **318** multiplied by the pressure of the hydraulic fluid. In various embodiments, the force that is generated to the head of piston **318** is reduced by the hydraulic fluid pressure at the tail end **322** of the piston **318**, which is return pressure via conduit **324** tied to hydraulic fluid flow source **332** via fourth conduit **336**, which, for example, may be at 50 psi (3.44738 bar). In that regard, the tail end **322** of the piston **318** is fluidly coupled to hydraulic fluid flow source **332** via fourth conduit **336**.

(16) Thus, in various embodiments, the higher hydraulic fluid pressure provided on the head end **314** of the piston **318** allows the piston **318** to translate in a second direction away from the head-end side **314** of the displacement intensifier **306** and increase the hydraulic fluid pressure to the hydraulic motor **250** via conduit **309**, the solenoid-operated directional control (or servo) valve **302**, and conduit **310**. For example, if the hydraulic fluid pressure on the head-end side **314** of the piston **318** is 2500 psi (1723689 bar) and there is a 2-to-1 area ratio between the head-end side **314** of the piston **318** and the back-end side **320** of the piston **318**, then a hydraulic fluid pressure of 5000 psi (3447379 bar) may be experienced on the back-end side **320** of the piston **318**. In keeping with the example, the 5000 psi (3447379 bar) of hydraulic fluid pressure increases the hydraulic fluid pressure to the hydraulic motor **250** via conduit **309**, the solenoid-operated directional control (or servo) valve **302**, and conduit **310** thereby overcoming 3500 psi (241.3165 bar) starting torque needed for the hydraulic motor **250** to rotate.

(17) In operation, once the hydraulic motor **250** starts to move, for example, below two RPMs, for example, one or two RPMs, the hydraulic fluid pressure required to steer the nose wheel assembly **118** will plummet because the torque efficiency suddenly increases by, for example, 15-20%. Thus, the hydraulic fluid pressure at the hydraulic motor **250** drops, which is also realized at the intensifier valve **304** via conduit **312**, which causes the intensifier valve **304** to re-center, and with reference to FIG. 3B, allows the hydraulic fluid to flow to the hydraulic motor **250** as described previously with regard to FIG. 3B.

(18) With additional reference to FIG. 3D, the process is reversed to enable turning the aircraft **100** in an opposite direction. In various embodiments, in operation, when a command is received from pilot steering input **120** via steering controller **330** to steer the nose landing gear assembly **106** to, for example, the left 20 degrees, 20 degrees per second, the command causes the solenoid-operated directional control (or servo) valve **302** to shift from a center flow-blocking position to a straight-flow position so that hydraulic fluid, flowing from the hydraulic fluid flow source **332** via third conduit **334**, flows through intensifier valve **304** (in a center straight-flow position) to the solenoid-operated directional control (or servo) valve **302** via conduit **308**. In that regard, the intensifier valve **304** is fluidly coupled to the hydraulic fluid flow source **332** via third conduit **334** and the

intensifier valve **304** is fluidly coupled to the solenoid-operated directional control (or servo) valve **302** via conduit **308**. It is noted that the hydraulic fluid also flows to a back-end side **320** of the displacement intensifier **306** via conduit **309** causing the piston **318** to translate in a first direction toward a head end **314** of the displacement intensifier **306** if not already bottomed on the head end **314** of the displacement intensifier's **306** cylinder. In that regard, the intensifier valve **304** is fluidly coupled to the back-end side **320** of the displacement intensifier **306** via conduit **309**.

(19) Accordingly, in various embodiments, the hydraulic fluid then flows through solenoid-operated directional control (or servo) valve **302**, which is the straight-flow position, to port **313** of the hydraulic motor **250** via conduit **326**, which fluidly couples the port **313** of the hydraulic motor **250** to the solenoid-operated directional control (or servo) valve **302**. In various embodiments, hydraulic fluid returning from hydraulic motor **250** flows to solenoid-operated directional control (or servo) valve **302** via conduit **310**, which fluidly couples the port **311** of the hydraulic motor **250** to the solenoid-operated directional control (or servo) valve **302**. The solenoid-operated directional control (or servo) valve **302**, which is in the straight-flow position, allows the hydraulic fluid to flow to the hydraulic fluid flow source **332** via fourth conduit **336**, which fluidly couples the solenoid-operated directional control (or servo) valve **302** to the hydraulic fluid flow source **332**.

(20) However, in various embodiments, the pressure of the hydraulic fluid, for example 2500 psi (172.3689 bar), delivered to the hydraulic motor **250** may not be enough to overcome a starting torque needed for the hydraulic motor **250** to rotate, for example 3500 psi (241.3165 bar). In various embodiments, in order to provide the necessary hydraulic fluid pressure to the hydraulic motor **250**, displacement intensifier **306** is engaged. In various embodiments, the hydraulic fluid provided to the hydraulic motor **250** is also provided to the intensifier valve **304** via conduit **315**, which is coupled to conduit **326**. In that regard, the solenoid-operated directional control (or servo) valve **302** is fluidly coupled to the control of the intensifier valve **304** via conduit **315**. In various embodiments, with reference to FIG. 3E, while the hydraulic fluid pressure is not enough to overcome the starting torque needed for the hydraulic motor **250** to rotate, the same hydraulic fluid pressure is enough to cause intensifier valve **304** to shift to the left, which provides for the hydraulic fluid flowing from the hydraulic fluid flow source **332** via third conduit **334** to flow through intensifier valve **304** (in a displacement-flow position) to a head-end side **314** of the displacement intensifier **306** via conduit **316**. In that regard, the intensifier valve **304** (in a displacement-flow position) is fluidly coupled to the head-end side **314** of the displacement intensifier **306** via conduit **316**.

(21) In various embodiments, a force is generated to the head of piston **318** that is proportional to the area of the head of the piston times the pressure of the hydraulic fluid. In various embodiments, that force that is generated to the head of piston **318** is reduced by the hydraulic fluid pressure at the tail end **322** of the piston **318**, which is return pressure via conduit **324** tied to hydraulic fluid flow source **332** via fourth conduit **336**, which, for example, may be at 50 psi (3.44738 bar). In that regard, the tail end **322** of the piston **318** is fluidly coupled to hydraulic fluid flow source **332** via fourth conduit **336**.

(22) Thus, in various embodiments, the higher hydraulic fluid pressure provided on the head end **314** of the piston **318** allows the piston **318** to translate in a second direction away from the head-end side **314** of the displacement intensifier **306** and increase the hydraulic fluid pressure to the hydraulic motor **250** via conduit **309**, the solenoid-operated directional control (or servo) valve **302**, and conduit **326**. For example, if the hydraulic fluid pressure on the head-end side **314** of the piston **318** is 2500 psi (172.3689 bar) and there is a 2-to-1 area ratio between the head-end side **314** of the piston **318** and the back-end side **320** of the piston **318**, then a hydraulic fluid pressure of 5000 psi (344.7379 bar) may be experienced on the back-end side **320** of the piston **318**. In keeping with the example, the 5000 psi (344.7379 bar) of hydraulic fluid pressure increases the hydraulic fluid pressure to the hydraulic motor **250** via conduit **309**, the solenoid-operated directional control (or servo) valve **302**, and conduit **326** thereby overcoming 3500 psi (241.3165 bar) starting torque

needed for the hydraulic motor **250** to rotate.

(23) In operation, once the hydraulic motor **250** starts to move, for example, one or two RPMs, the hydraulic fluid pressure required to steer the nose wheel assembly **118** will plummet because the torque efficiency suddenly jumps up by, for example, 15-20%. Thus, the hydraulic fluid pressure at the hydraulic motor **250** drops which is also realized at the intensifier valve **304** via conduit **315**, which causes the intensifier valve **304** to re-center, and with reference to FIG. 3D, allows the hydraulic fluid to flow to the hydraulic motor **250** as described previously with regard to FIG. 3D.

(24) Thus, in various embodiments, by providing a hydraulic motor start assist mechanism as described, the bi-directional hydraulic motor hydraulic motor may be sized approximately 20-30% smaller compared to what the bi-directional hydraulic motor hydraulic motor would normally be sized at without the hydraulic motor start assist mechanism. In various embodiments, the bi-directional hydraulic motor hydraulic motor weight savings of approximately 25% is reduced to approximately 15-20% by the added weight of the integral intensifier and intensifier valve of the hydraulic motor start assist mechanism.

(25) Benefits and other advantages have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C.

(26) Systems, methods, and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(27) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is intended to invoke 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

Claims

1. A nose-wheel steering system, comprising: a hydraulic motor including a drive shaft configured to rotate about a first axis; and a hydraulic motor start assist mechanism coupled to the hydraulic motor and configured to provide a temporary boost of hydraulic fluid pressure supplied to the hydraulic motor, wherein the hydraulic motor start assist mechanism comprises: a servo valve coupled to the hydraulic motor and an intensifier valve, the servo valve configured to provide the hydraulic fluid pressure to the hydraulic motor and the intensifier valve; the intensifier valve coupled to a displacement intensifier and configured to provide the hydraulic fluid pressure to the servo valve in a first state and provide the hydraulic fluid pressure to the displacement intensifier in a second state; and the displacement intensifier configured to translate in a first direction in the first state and translate in a second direction in the second state.
2. The nose-wheel steering system of claim 1, wherein: responsive to receiving a first command from a pilot steering input, the servo valve switches from a flow-blocking position to a cross-flow position, in the cross-flow position, the servo valve provides pressurized hydraulic fluid to a first port on the hydraulic motor, and responsive to the first command being discontinued, the servo valve switches from the cross-flow position to the flow-blocking position.
3. The nose-wheel steering system of claim 2, further comprising: responsive to the hydraulic fluid pressure at the first port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, a rise in the hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the first port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor.
4. The nose-wheel steering system of claim 3, wherein an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in the hydraulic fluid pressure at the first port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement intensifier.
5. The nose-wheel steering system of claim 3, wherein the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by the area of the head end of the displacement intensifier being greater than an area at a tail end the displacement intensifier thereby allowing a piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at a back end of the displacement intensifier to increase in pressure.
6. The nose-wheel steering system of claim 1, wherein: responsive to receiving a second command from a pilot steering input, the servo valve switches from a flow-blocking position to a straight-flow position, in the straight-flow position, the servo valve provides pressurized hydraulic fluid to a second port on the hydraulic motor, and responsive to the second command being discontinued, the servo valve switches from the straight-flow position to the flow-blocking position.
7. The nose-wheel steering system of claim 6, further comprising: responsive to the hydraulic fluid pressure at the second port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, the hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the second port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor.
8. The nose-wheel steering system of claim 7, wherein an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in the hydraulic fluid pressure at the second port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement intensifier.
9. The nose-wheel steering system of claim 7, wherein the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by the area at the head end of the displacement intensifier being greater than an area at a tail end of the displacement intensifier thereby allowing a

piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at a back end of the displacement intensifier to increase in pressure.

10. A nose landing gear assembly, comprising: a shock strut assembly including a strut cylinder and a strut piston configured to telescope relative to the strut cylinder; and a nose-wheel steering system coupled to the shock strut assembly and configured to rotate the strut piston about a piston axis of rotation, the nose-wheel steering system comprising: a hydraulic motor including a drive shaft configured to rotate about a second axis, the second axis being parallel to the piston axis of rotation; and a hydraulic motor start assist mechanism coupled to the hydraulic motor and configured to provide a temporary boost of hydraulic fluid pressure supplied to the hydraulic motor, wherein the hydraulic motor start assist mechanism comprises: a servo valve coupled to the hydraulic motor and an intensifier valve, the servo valve configured to provide the hydraulic fluid pressure to the hydraulic motor and the intensifier valve; the intensifier valve coupled to a displacement intensifier and configured to provide the hydraulic fluid pressure to the servo valve in a first state and provide the hydraulic fluid pressure to the displacement intensifier in a second state; and the displacement intensifier configured to translate in a first direction in the first state and translate in a second direction in the second state.

11. The nose landing gear assembly of claim 10, wherein: responsive to receiving a first command from a pilot steering input, the servo valve switches from a flow-blocking position to a cross-flow position, in the cross-flow position, the servo valve provides pressurized hydraulic fluid to a first port on the hydraulic motor, and responsive to the first command being discontinued, the servo valve switches from the cross-flow position to the flow-blocking position.

12. The nose landing gear assembly of claim 11, further comprising: responsive to the hydraulic fluid pressure at the first port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, a rise in the hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the first port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor.

13. The nose landing gear assembly of claim 12, wherein an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in the hydraulic fluid pressure at the first port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement intensifier, and wherein the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by the area at the head end of the displacement intensifier being greater than an area at a tail end of the displacement intensifier thereby allowing a piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at the back end of the displacement intensifier to increase in pressure.

14. The nose landing gear assembly of claim 10, wherein: responsive to receiving a second command from a pilot steering input, the servo valve switches from a flow-blocking position to a straight-flow position, in the straight-flow position, the servo valve provides pressurized hydraulic fluid to a second port on the hydraulic motor, and responsive to the second command being discontinued, the servo valve switches from the straight-flow position to the flow-blocking position.

15. The nose landing gear assembly of claim 14, further comprising: responsive to the hydraulic fluid pressure at the second port on the hydraulic motor failing to overcome a starting torque of the hydraulic motor, the hydraulic fluid pressure causes the intensifier valve to switch from a straight-flow position to a displacement-flow position providing the pressurized hydraulic fluid to a head end of the displacement intensifier thereby causing the hydraulic fluid pressure at the second port on the hydraulic motor to increase to overcome the starting torque of the hydraulic motor.

16. The nose landing gear assembly of claim 15, wherein an area ratio between the head end of the displacement intensifier and a back end of the displacement intensifier provides for the increase in

the hydraulic fluid pressure at the second port on the hydraulic motor when an area at the head end of the displacement intensifier is greater than an area at the back end of the displacement intensifier, and wherein the increase in the hydraulic fluid pressure provided by the displacement intensifier is caused by the area at the head end of the displacement intensifier being greater than an area at a tail end of the displacement intensifier thereby allowing a piston in the displacement intensifier to translate in the first direction and cause the hydraulic fluid pressure at the back end of the displacement intensifier to increase in pressure.

17. A shock strut assembly for an aircraft landing gear assembly, comprising: a strut cylinder; a strut piston configured to telescope relative to the strut cylinder; and a steering system coupled to the strut piston and configured to rotate the strut piston about a piston axis of rotation, the steering system comprising: a hydraulic motor including a drive shaft configured to rotate about a second axis, the second axis being parallel to the piston axis of rotation; and a hydraulic motor start assist mechanism coupled to the hydraulic motor and configured to provide a temporary boost of hydraulic fluid pressure supplied to the hydraulic motor, wherein the hydraulic motor start assist mechanism comprises: a servo valve coupled to the hydraulic motor and an intensifier valve, the servo valve configured to provide the hydraulic fluid pressure to the hydraulic motor and the intensifier valve; the intensifier valve coupled to a displacement intensifier and configured to provide the hydraulic fluid pressure to the servo valve in a first state and provide the hydraulic fluid pressure to the displacement intensifier in a second state; and the displacement intensifier configured to translate in a first direction in the first state and translate in a second direction in the second state.
